🔷 16.11.6 Evaluation and Decision Scoring Integration (Final — Integrated Version)

Instantaneous Evaluation Function

$$f_i(t) = \overline{\mathcal{E}}(\Psi_i(t))\,\nabla \Phi(\Psi_i(t)) - 2\sigma^2 \nabla \ln P_i(t)$$

Under compactness of Ω and bounded gradients:

$$|f_i(t)| \le C_f$$

Windowed Evaluation

$$G_{i,u}^k = \int_{t_k}^{t_k+\Delta T} f_{i,u}(t)\, e^{-\lambda (t-t_k)} dt$$

Thus:

$$|G_{i,u}^k| \le \frac{C_f}{\lambda}$$

Normalization

$$
\tilde{G}_{i,u}^k =
\frac{G_{i,u}^k}{E_{\mathrm{ref}}}, \quad
E_{\mathrm{ref}} > 0
$$

Thus:

$$
|\tilde{G}_{i,u}^k| \le C
$$

PAC-Based Weighting

$$
W_i^k =
\frac{PAC_i^k}{\sum_j PAC_j^k + \varepsilon_w},
\quad
W_i^k \in (0,1), \quad \sum_i W_i^k = 1
$$

Thus:

decision aggregation is a convex combination.

Decision Score

$$
D_u(k) =
\sum_i W_i^k\, \tilde{G}_{i,u}^k
$$

Thus:

$$|D_u(k)| \le C_D$$

Temporal Smoothness

$$|D_u(k+1) - D_u(k)| \le L_D\,dt$$

Thus:

evaluation evolves smoothly without abrupt jumps.

Goal Alignment

$$f_i \supset \nabla \Phi(\Psi_i)$$

Thus:

scores are aligned with the system objective.

Real-Time Consistency

$$G_{i,u}^k \text{ is computed over } [t_k, t_k+$$

\Delta T]

Thus:

no delay mismatch occurs between state, probability, and evaluation.

Stability Compatibility

Since:

- f_i is bounded
- weights W_i^k are bounded and normalized
- scores D_u(k) are bounded

Thus:

the evaluation layer preserves Lyapunov stability and introduces no destabilizing effects.

Implication

\boxed{
\text{The evaluation layer produces bounded, smooth, and objective-aligned decision scores,

consistent with closed-loop stability.}
}

SEKHEM-TSDD Theory, Mohamed Abd Elnaby,
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